COFFEE SHOP

Sales Analysis

Python Analysis & Power BI Dashboard

Project Documentation

Dataset period: April 2019

1. Project Overview

This project analyzes sample coffee shop sales and pastry inventory data to understand sales performance, product contribution, customer-related sales patterns, and recorded pastry waste. The work combines Python for data inspection and analysis with Power BI for interactive reporting.

The objective is to turn the available operational data into clear business insights while documenting data quality issues and limitations that affect interpretation.

2. Dataset Source and Scope

Source: Kaggle — Coffee shop sample data (11.1.3+), published by ylchang.

Dataset URL: https://www.kaggle.com/datasets/ylchang/coffee-shop-sample-data-1113

The workbook used in the analysis contains multiple related sheets, including sales receipts, products, customers, staff, sales outlets, dates, sales targets, generations, and pastry inventory.

Data area	Role in the project
Sales receipts	Transaction-level sales, quantities, prices, dates, times, outlets, and product IDs.
Product	Product names, groups, categories, types, and price-related attributes.
Customer / Generations	Customer attributes and generation categories used for customer-related analysis.
Sales outlet / Staff	Outlet and staff reference information.
Dates	Calendar attributes for reporting by date.
Pastry inventory	Opening pastry quantities, quantities sold, and recorded waste for selected outlets.
Sales targets	Monthly outlet targets included in the source workbook; not used as a main comparison in the current dashboard.

The sales receipts cover April 2019. The pastry inventory records in this workbook cover outlets 3, 5, and 8. Therefore, pastry waste findings should not be generalized to every product or every outlet.

3. Tools and Technologies

- Python — data loading, inspection, quality checks, aggregation, and exploratory analysis.
- Pandas and NumPy — data manipulation and numerical analysis.
- Matplotlib and Seaborn — exploratory charts and distribution review.
- Power BI — data model, DAX measures, interactive visuals, and dashboard reporting.
- Jupyter Notebook — reproducible analysis workflow.

4. Project Workflow

4.1 Data loading and initial inspection

The workbook sheets were loaded into pandas DataFrames. The analysis inspected dataset dimensions, column names, data types, missing values, duplicate records, and unique key values.

```
import pandas as pd
from pathlib import Path

file = Path("converted.xlsx")
sales = pd.read_excel(file, sheet_name="201904 sales reciepts")
product = pd.read_excel(file, sheet_name="product")
```

The relative file path makes the notebook easier to reuse: the Excel workbook should be placed in the same folder as the notebook before running it.

4.2 Data quality and relationship checks

Reference-table identifiers were checked for uniqueness, and sales identifiers were compared with their corresponding reference tables. The sales data contains customer IDs that do not match a known customer record, including customer_id = 0. These records were retained rather than removed.

4.3 Sales and product analysis

Sales were joined to the product reference table using product_id. Revenue was then summarized by product group and product to identify the largest contributors to sales.

```
sales_merged = pd.merge(
    sales,
    product,
    on="product_id",
    how="left"
)

sales_by_group = (
    sales_merged.groupby("product_group")["line_item_amount"]
    .sum()
    .sort_values(ascending=False)
)
```

4.4 Time-based analysis

Transaction time was converted into an hour field to compare sales activity throughout the day. The transaction date was used for daily sales trends. Since the sales data covers one month, the analysis does not claim a recurring weekday or seasonal pattern.

```
sales["hour"] = pd.to_datetime(
    sales["transaction_time"],
    format="%H:%M:%S"
).dt.hour
```

4.5 Pastry waste analysis

Pastry inventory was analyzed separately from general sales. Recorded waste was compared with start-of-day quantity. The analysis uses the recorded waste field and does not assume that every unsold item was wasted.

```
total_waste = pastry_inventory["waste"].sum()
total_start = pastry_inventory["start_of_day"].sum()
waste_rate = total_waste / total_start
```

4.6 Outlier review

Potential outliers were reviewed as part of data understanding. They were not automatically removed from the primary analysis because unusually large sales values may be valid transactions. No business evidence was available to classify those values as errors.

5. Key Findings

Area	Finding	Interpretation
Overall sales	$233,635.95	Total recorded sales revenue for the April 2019 sales receipts.
Product groups	Beverages: 77.29%	Beverages are the largest contributor to recorded revenue.
Product groups	Beverages + Food: 88.82%	Most recorded revenue comes from these two groups.
Time of day	10:00 AM	Highest revenue hour in the analyzed data.
Transaction IDs	11:00 AM	Highest count of distinct transaction_id values by hour; this is an ID-based measure.
Customer matching	$119,312.72 (51.07%)	Revenue associated with customer_id = 0; this is a share of revenue, not a share of customers.
Pastry inventory	4,500 recorded waste units	Sum of the waste field in the available pastry inventory records.
Pastry inventory	60.93%	Recorded waste divided by start-of-day quantity (4,500 / 7,386).
Waste by product	Ginger Scone: 2,140 units	Largest recorded waste quantity; 47.56% of total recorded pastry waste.
Waste by outlet	Outlet 5: 63.10%	Highest overall recorded waste rate among outlets 3, 5, and 8.

6. Business Recommendations

- Review beverage operations and product availability because beverages contribute the largest share of revenue.
- Use the 10:00 AM revenue peak as a starting point for reviewing staffing and preparation coverage; validate with additional months before changing schedules.
- Investigate Ginger Scone production and demand patterns, since it has the highest recorded waste quantity.
- Review pastry preparation and recording practices at outlet 5, which has the highest recorded waste rate among the three outlets with inventory data.
- Improve customer ID capture and matching so that more sales can be connected to known customer records.

These are investigation priorities, not proven causes. Operational changes should be based on additional periods of data and discussions with store teams.

7. Power BI Dashboard

The Power BI report is organized into two pages. The first page provides a sales overview; the second page combines product, pastry waste, and customer-related analysis.

7.1 Sales Overview

- KPI cards: Total Sales, Total Transactions, Total Quantity, and Average Transaction Value.
- Sales by Product Group, including the contribution of each group.
- Sales distribution by outlet.
- Sales by hour and daily sales trend.
- Top products by sales.



7.2 Product & Customer Analysis

- Total recorded waste by product.
- Product waste rate by product.
- Waste rate by outlet.
- Sales by generation and sales by gender.
- KPI cards for pastry inventory and sales associated with customer_id = 0.



8. Data Limitations and Interpretation Notes

- The sales receipts represent April 2019 only; results may not reflect other months or seasons.
- Pastry inventory data is available for outlets 3, 5, and 8 only, and it relates to pastry inventory rather than all products.
- A large share of revenue is associated with customer_id = 0. This does not mean that the same share of customers is unidentified.
- Some customer IDs in sales do not match records in the customer reference table.
- The sum of sold and wasted pastry units is 7,354, compared with 7,386 start-of-day units, leaving a 32-unit difference. The available data does not establish the reason.
- Some inventory rows show quantities that do not reconcile directly. These were retained and treated as data limitations rather than silently corrected.
- Potential outliers were not removed without evidence that they were invalid.
- The meaning of every coded category is not documented in the source data; values such as gender = 'N' should not be assigned an assumed meaning.

9. Conclusion

The project demonstrates an end-to-end workflow for exploring coffee shop sales and selected pastry inventory data. Python was used to inspect and analyze the source workbook, while Power BI was used to present the findings in an interactive dashboard. The analysis highlights the importance of documenting data limitations and avoiding unsupported assumptions when translating data into business recommendations.

10. Reproducibility and Project Files

Recommended project folder structure:

```
Coffee-Shop-Sales-Analysis/
├── README.md
├── Coffee_Shop_Sales_Analysis.ipynb
├── Coffee Shop Sales Analysis.pbix
├── Project Documentation.docx
├── converted.xlsx
└── screenshots/
    ├── sales-overview.png
    └── product-customer-analysis.png
```

To run the notebook, place converted.xlsx in the same folder as the notebook, then run all cells from top to bottom. If the Excel workbook is not redistributed in the repository, provide the Kaggle source link and explain how to obtain it.

Before redistributing the original workbook, check the dataset's license and usage terms on Kaggle. The source link and creator attribution should remain in the README and project documentation.

11. Dataset Attribution

Dataset title: Coffee shop sample data (11.1.3+)

Publisher/uploader: ylchang

Platform: Kaggle

Source: https://www.kaggle.com/datasets/ylchang/coffee-shop-sample-data-1113

This project is an independent analysis of the referenced sample dataset. All findings are based on the workbook version used for this project.